Enhancing the Kingdom of Saudi Arabia as the world leader in clinical research data publishing

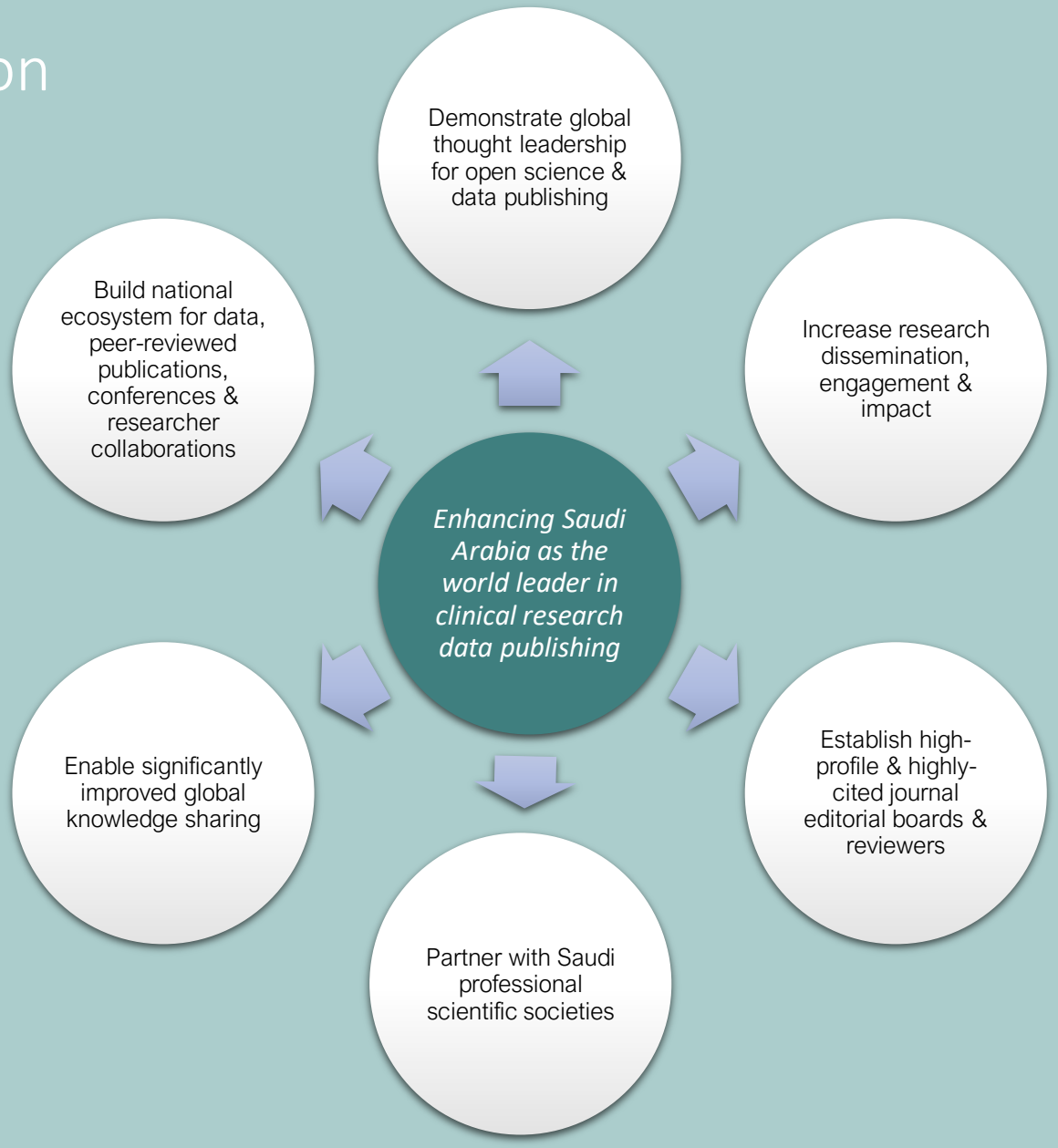
In support of

VISION رؤيـــة 2030

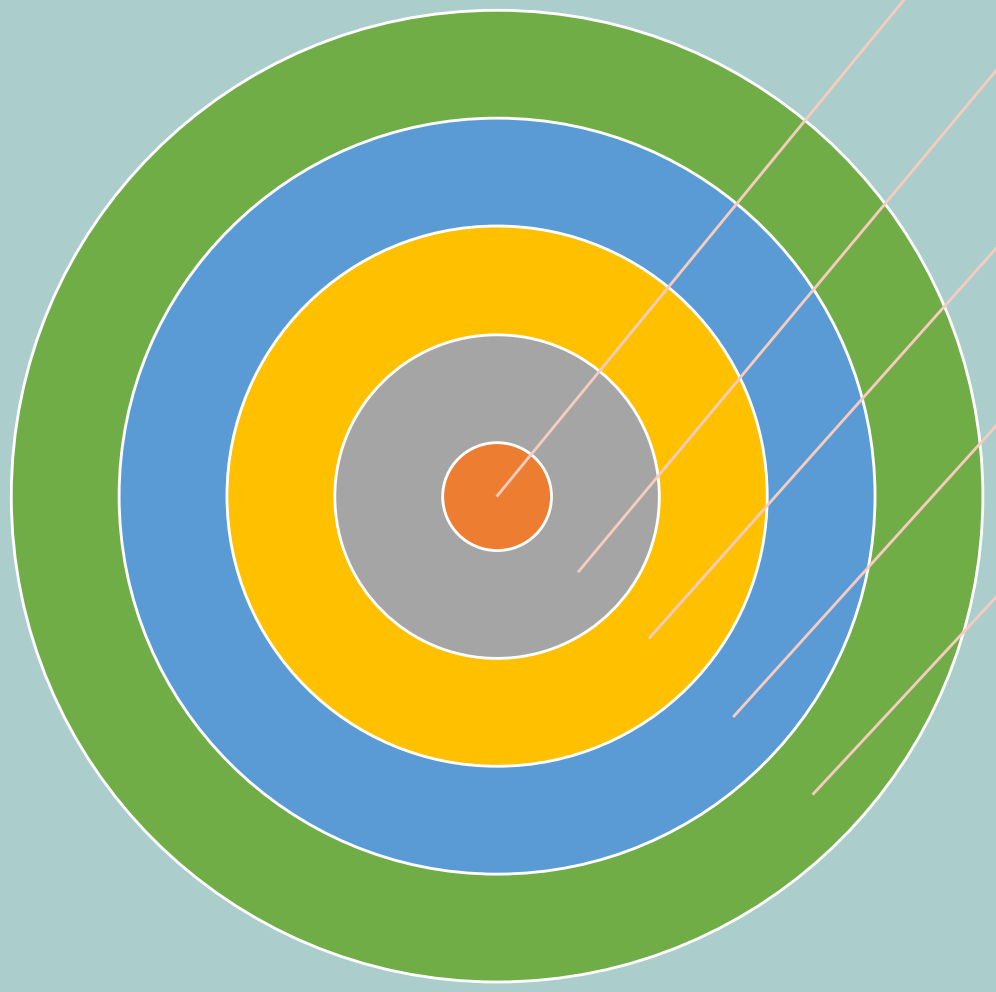
المملكة العربية السعودية

KINGDOM OF SAUDI ARABIA

1. Vision & mission



2. Goals



Build modern, open-source enabled publication platforms which are interoperable with all relevant national & international databases

Deliver simplified and single-sign on national access to evidence-based knowledge resources

Increase number, quality & international impact of research publications, to include developing existing Saudi journals

By publishing all data under international open science principles, increase patient & physician engagement

Guarantee high levels of research integrity, trust & reliability of Saudi research data

3. Problem to be solved

KNOWLEDGE SHARING

- Inefficient sharing of knowledge between researchers, practitioners & policy makers
- No unified national repository for access to research publications, data or abstracting & indexing services

INTERNATIONAL STATUS OF SAUDI RESEARCH

- Low international status & credibility
- Researchers lack international profiles, impact & rankings
- National journals do not have high citation impact & rankings

OPEN SCIENCE

- Currently, no national open science funding mandate
- Limited open access publishing expertise
- Majority of existing Saudi-affiliated journals not fully open access

RESEARCH INTEGRITY

- Increasing reputational risk for existing & future Saudi researchers who have published in non-Saudi owned journals (e.g., 3,748 Saudi-affiliated papers published between 2017-22 now negatively impacted by Hindawi & MPDI delisting from Clarivate Journal Citation Index)
- As a consequence, there is urgent need for greater editorial governance and focus on quality over quantity

4. Opportunity: global drivers for change

Increased focus on data

- exponential growth of data sets requiring hosting & curation

Digital transformation

- continued innovation in systems & metadata

Open science

- increasingly mandated by government research funders

Open source technologies

- enabling modern, multichannel, interoperable & low-cost processes

Post-COVID awareness

- knowledge sharing critical to advancing science

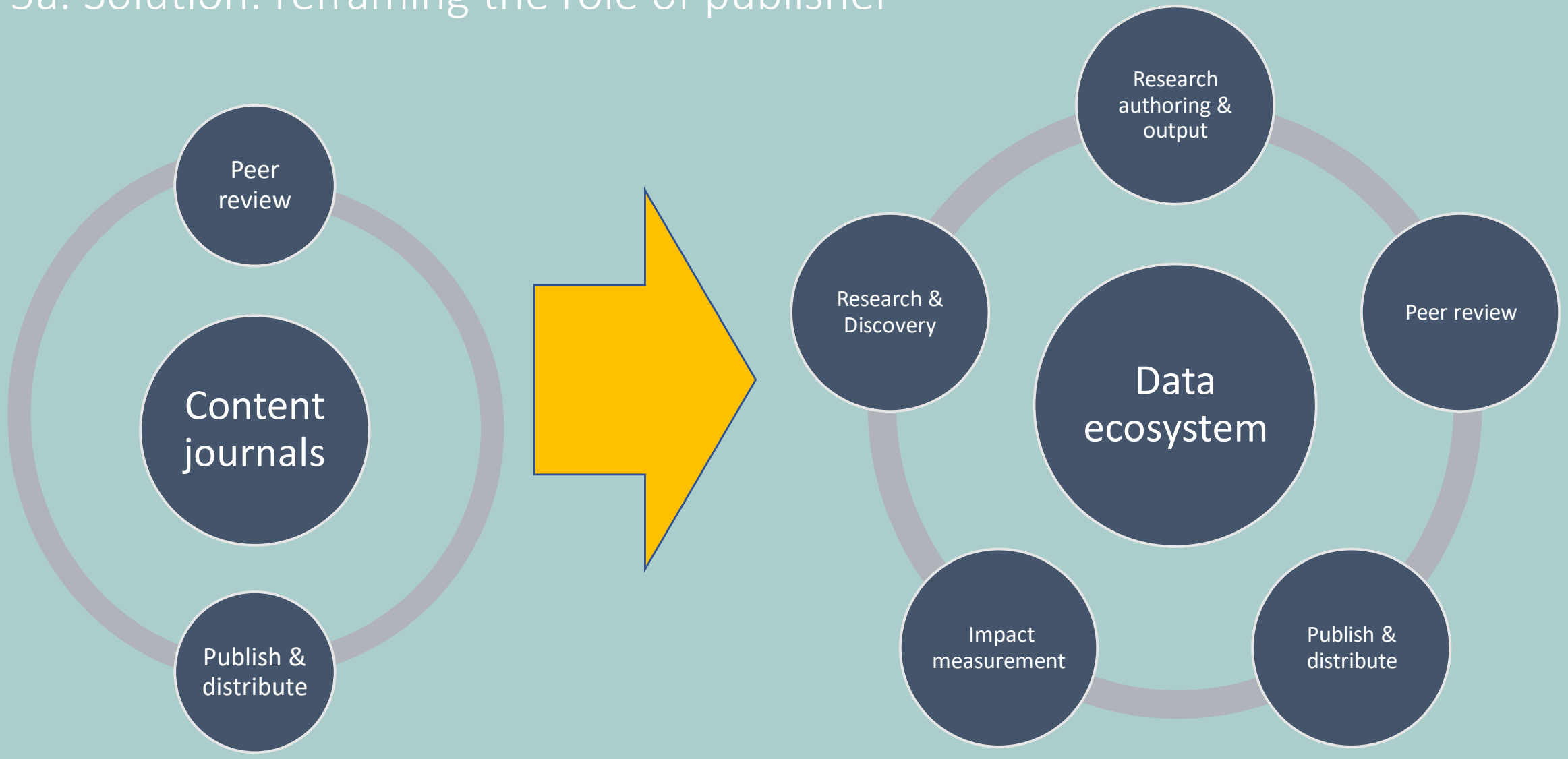
Multimedia

- increased use of video, voice & text content

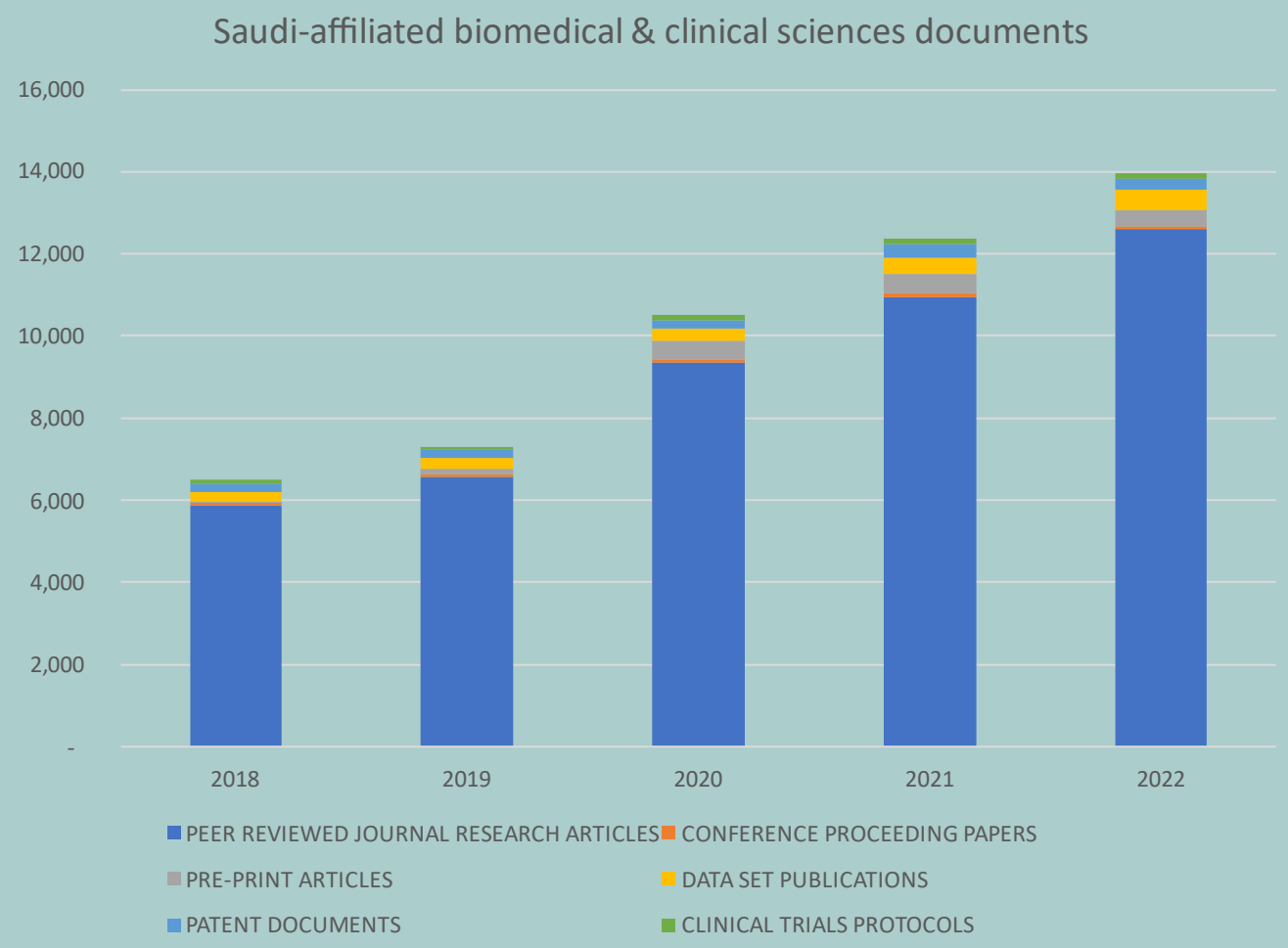
AI technologies

- exponential increase in use of ChatGPT & machine learning applications

5a. Solution: reframing the role of publisher



5b. Creation of unified database of Saudi-affiliated research



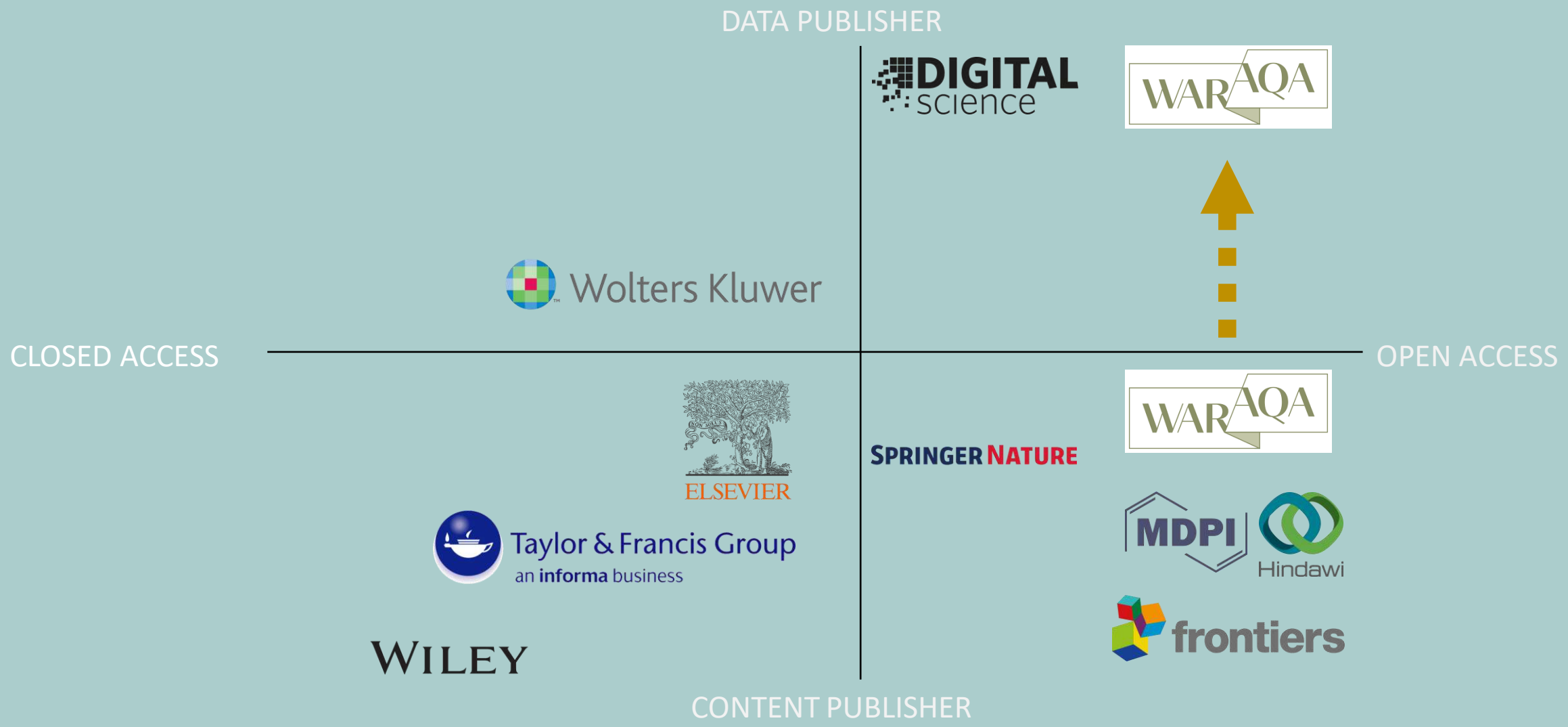
Notes:

- Biomedical & clinical sciences market
- 5-Y CAGR all articles: Saudi Arabia 21% vs Global 6%
- % all articles OA: Saudi Arabia 69% vs Global 48%
- Current OA market: **Saudi Arabia US$18.3m** vs Global US$1,264m
- Potential market if all Saudi articles published as OA: US$26.5m
- Waraqa targets 25% market share = US$6.6m

Data Sources:

- Dimensions (Digital Science)

5c. Reposition Waraqa to gain competitive advantage



6. Go to market - a unique multi-channel approach

CHANNEL	OUTPUT	CUSTOMER	TECHNOLOGY	COMPETITORS
1. NATIONAL DATA REPOSITORY	Research datasets, student theses	Government, University Hospital, PhD student	Dspace, ORCID	Saudi Digital Library
2. PRE-PRINT SERVER	Research publications (not peer reviewed)	Researchers	Figshare Preprints	BioArxiv, MedArXiv
3. ACADEMIC JOURNALS	Research publications (peer reviewed)	Researchers	Publons, Kotahi,Crossref, Altmetric, Dimensions	Hindawi, MDPI, Wolters Kluwer
4. EVIDENCE-BASED REVIEWS DATABASE	AI-curated article & research data	Clinicians, Patients	New build	UpToDate, Ovid, Cochrane Reviews, First Consult

7a. Challenges & solutions

Financial

Challenge:

- Acquire open access article publication fees from individual authors
- Build subscription-based services

Solution:

- Centrally fund fees via universities
- Upsell subscription services via third-party vendors, e.g. ProQuest

Research Integrity

Challenge:

- To underpin Vision 2030, Saudi Arabia must commit to robust research integrity, to attract both Saudi and international authors

Solution:

- Establish editorial governance structure led by internationally respected Saudi academics

Technology

Challenge:

- Build services that are interoperable with customers' existing workflows and systems
- Enrich content with metadata and machine learning tools to maximise discovery and usage

Solution:

- Integrate existing and proven technologies adopted by universities, researchers and publishers
- Adopt modern and modular open source software solutions

Competition

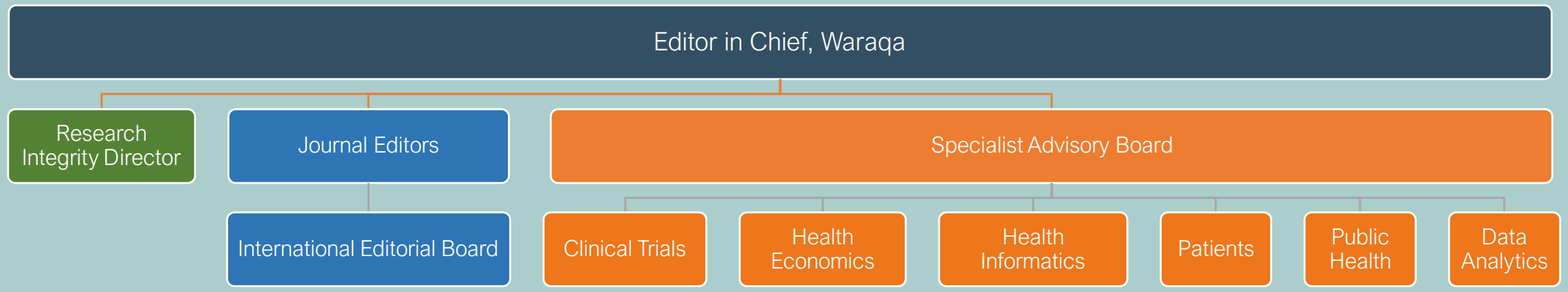
Challenge:

- Compete with established international publishers
- Increase participation from Saudi research authors
- Attract international authors

Solution:

- promote close alignment with and support of Vision 2030
- market to all Saudi-affiliated authors and research institutes
- partner with Saudi membership societies

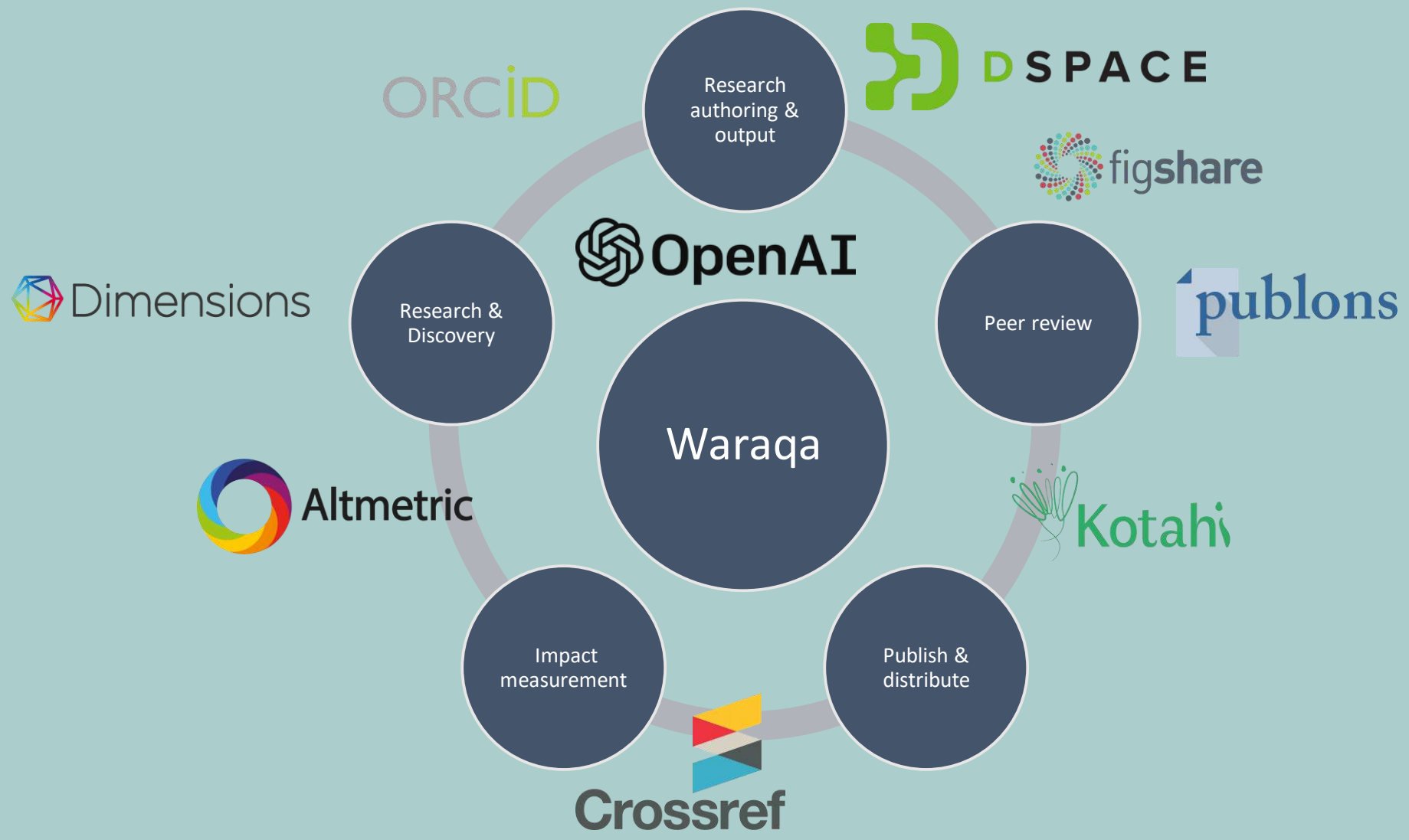
7b. Editorial governance structure



Notes:

- Editor-in-Chief has oversight off all research publications and data
- Research Integrity Director sets guidelines and controls for publication ethics
- Journal Editors are responsible for individual titles
- Specialist Advisory Board provides expertise across broad subject areas

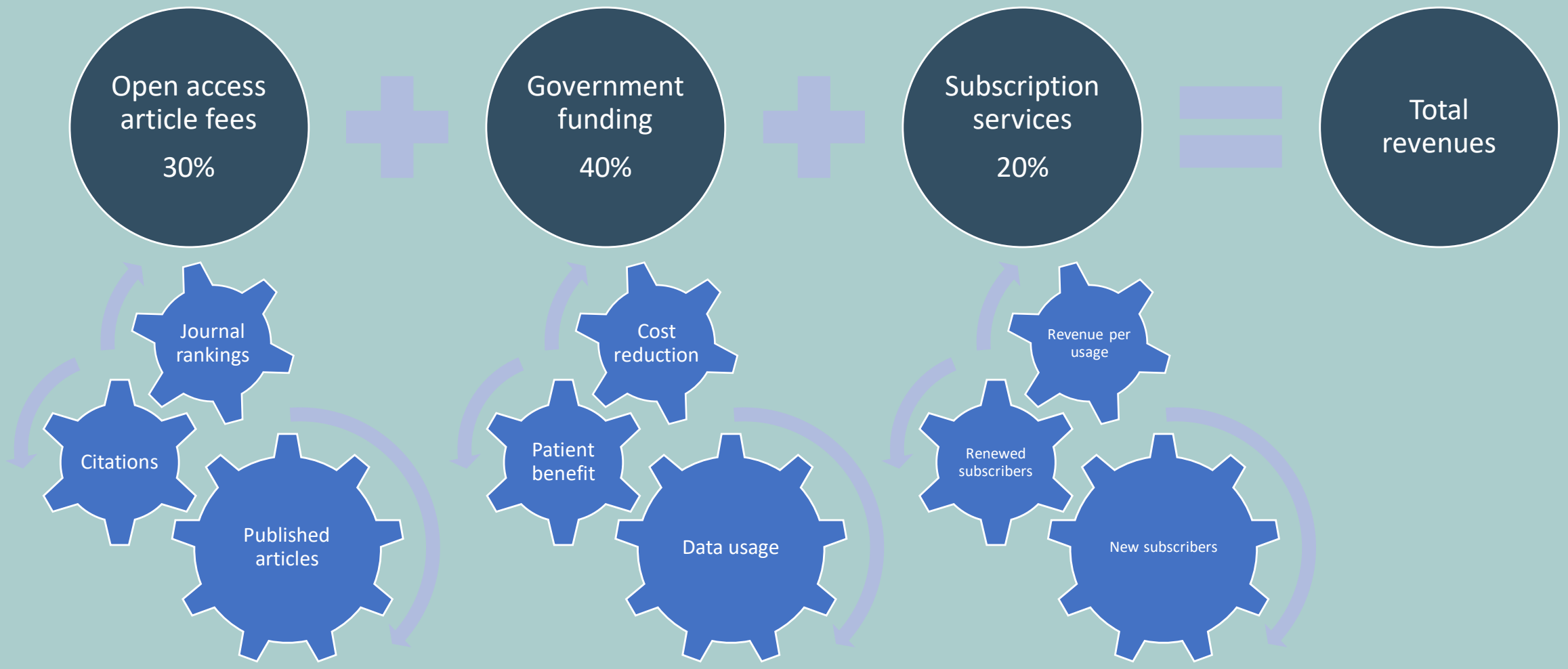
6a. Technical partnerships



6b. Data ecosystem components

COMPONENT	PURPOSE
DSPACE	Digital repository system for managing and preserving digital content such as documents, images, and videos. It is an open-source software platform that helps organizations to create and maintain institutional repositories.
figshare	Web-based platform for sharing and publishing research outputs, including figures, datasets, software, and other materials. It allows researchers to make their research outputs available in a citable, shareable, and discoverable manner.
publons	Allows researchers to track and showcase their publications, peer review contributions, and editorial work. It also provides a way for institutions and funders to track the research output of their staff or grantees.
Kotahi	Digital first, open-source and multichannel publishing ecosystem developed by the Coko Foundation. Encompasses all elements of submission, peer review, editing, typesetting and publishing platform.
Crossref	Makes scholarly communication more efficient, effective, and equitable. Works with publishers, libraries, researchers, and other stakeholders to ensure that scholarly content is easily discoverable, accessible and reusable.
Altmetric	Measures the impact of scholarly research outputs, such as articles and datasets, in a way that goes beyond traditional citation metrics. Captures the broader reach, attention, and influence of research outputs in the digital age, by looking at factors such as social media mentions, downloads, and views.
ORCID	Open Researcher and Contributor ID is a unique and persistent digital identifier, widely used in academic and research communities to support automated linkages between researchers and their professional activities, such as publications, grants, and patents.
Dimensions	Digital Science Dimensions is a research insights platform that allows users to explore and analyze research trends, funding data, and publication data. It covers various research areas such as life sciences, physical sciences, and social sciences.
OpenAI	OpenAI GPT (Generative Pre-trained Transformer) is an artificial intelligence model that uses deep learning to generate human-like text. It is used in natural language processing tasks such as language translation, text summarization, and chatbot development.

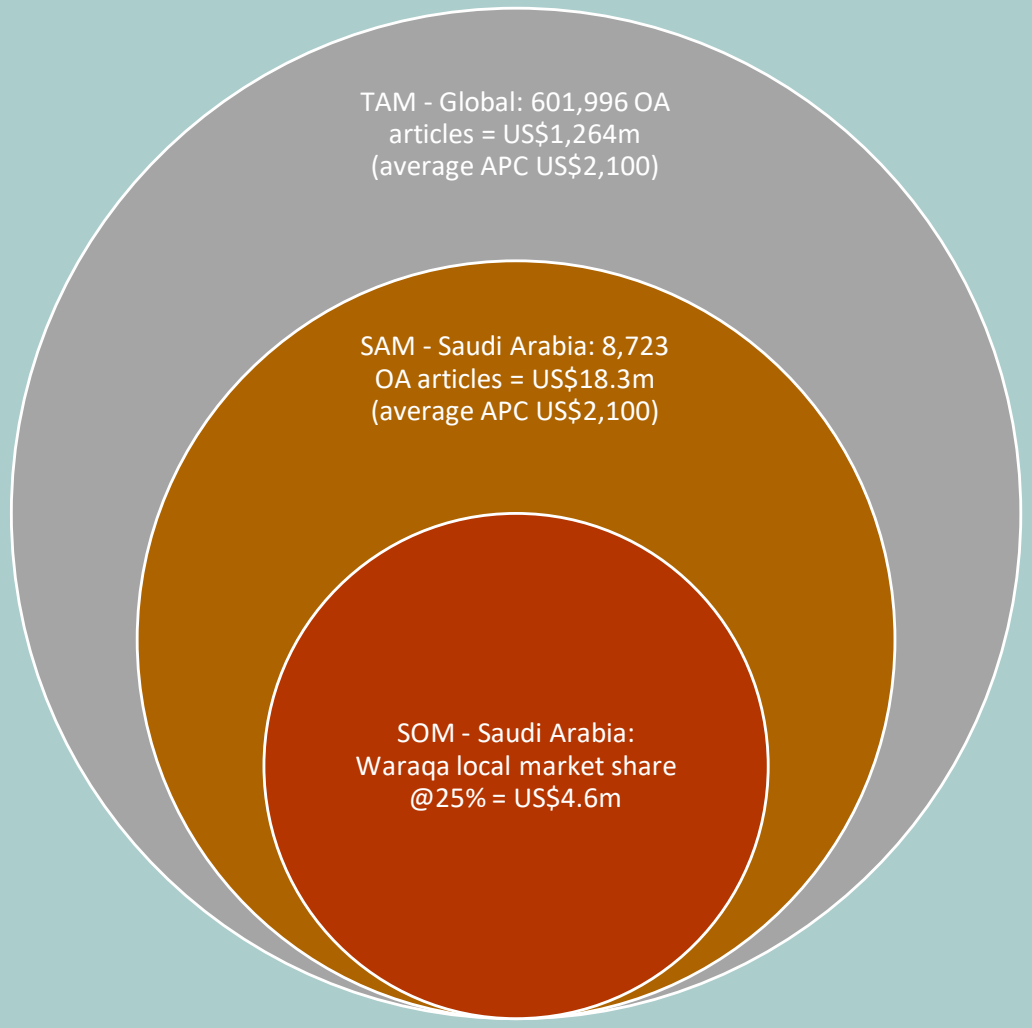
7. Revenue mix & KPIs

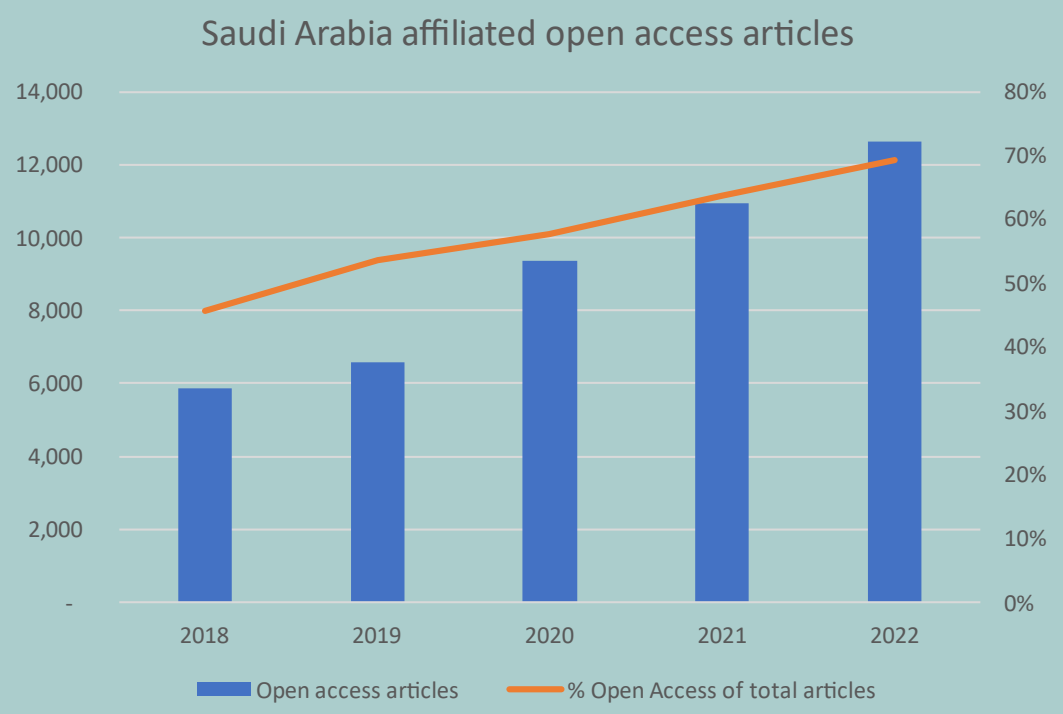


8. Project roadmap

	2026				2027			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
NATIONAL DATA REPOSITORY								
Build platform								
Launch								
PRE-PRINT SERVER								
Build platform								
Launch								
PEER-REVIEWED JOURNALS								
Build platform								
Transfer existing journals & acquire new								
EVIDENCE-BASED REVIEWS DATABASE								
Build platform								
Create content								

9a. Market size: current view





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Data Sources:

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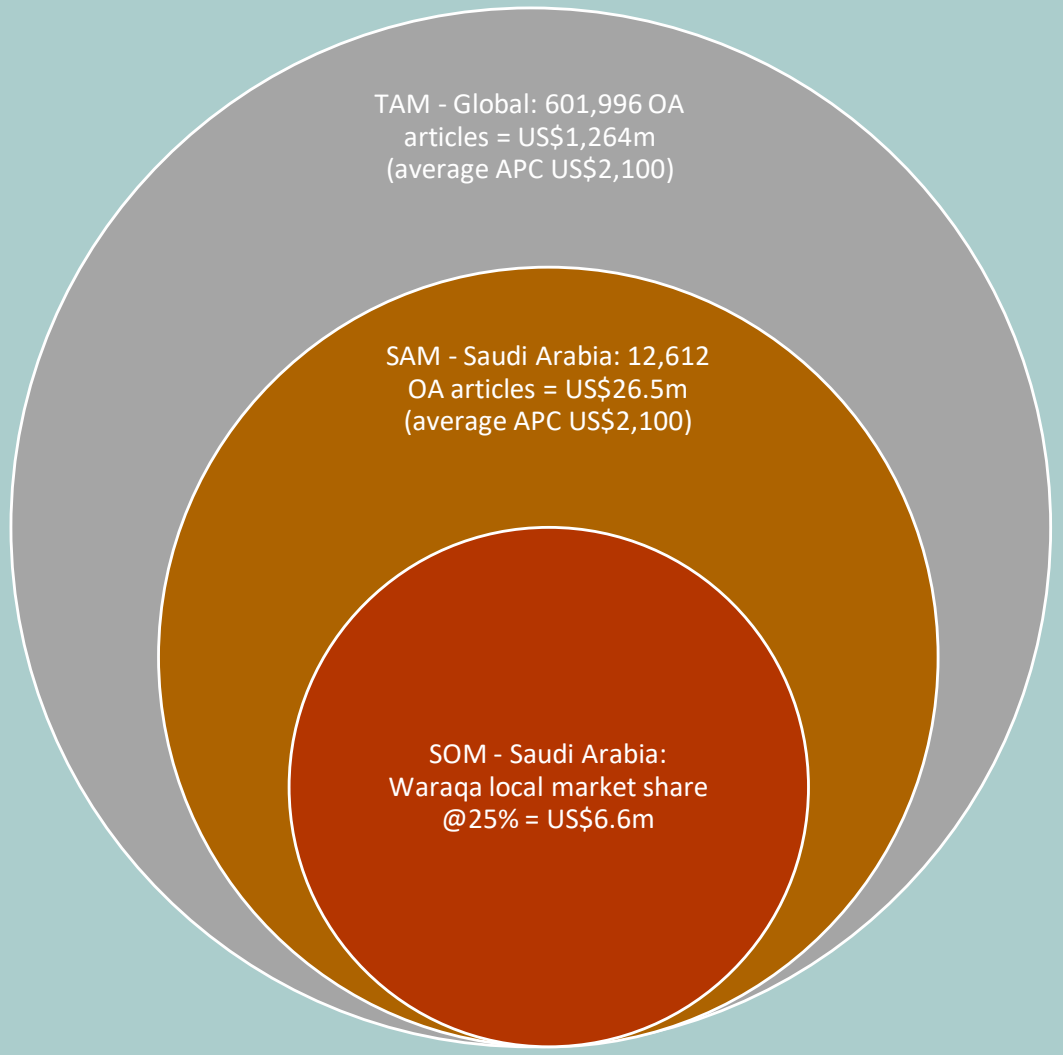
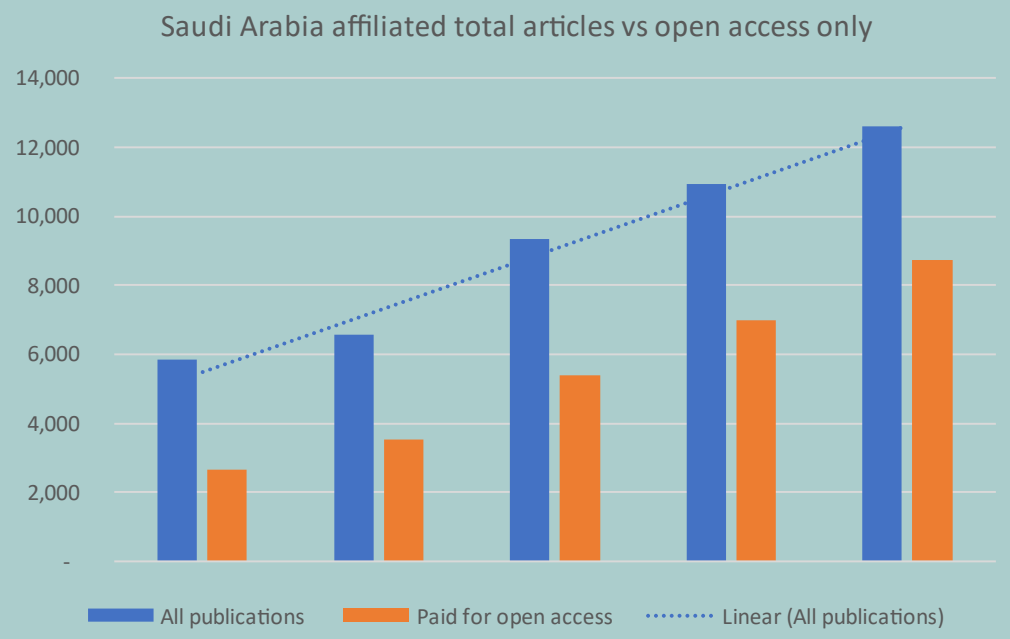
- *Article Processing Charges based publications: to which extent the price explains scientific impact?*, Jul 2021 https://doi.org/10.48550/arXiv.2107.07348

9b. Market size: future view





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